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Class: C-3
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TITLE: Title: Working with Geospatial data

Objective:

To analyze and interpret the visualization the geospatial Data

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Expected OUTCOME of Experiment: CO 1

Learn how to locate and download datasets, extract insights from that data and present their findings in a variety of different formats

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Books/ Journals/ Websites referred:

1. Data Analysis with Microsoft Excel, Wayne L. Winston, Microsoft Press.
2. Excel 2019 Bible, John Walkenbach, Wiley.
3. Practical Statistics for Data Scientists, Peter Bruce and Andrew Bruce, O'Reilly Media

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Background Theory:
Tableau provides comprehensive support for working with geospatial data, allowing users to visualize, analyze, and interpret spatial patterns in an intuitive and interactive manner. It accommodates various geographic data types, including latitude–longitude coordinates, administrative regions such as countries, states, cities, and postal codes, as well as custom spatial datasets like shapefiles and GeoJSON files. These features enable practical real-world applications such as visualizing water quality parameters across river networks, identifying pollution-prone zones along coastlines, and assessing spatial variations in environmental indicators across regions and time periods. In urban and infrastructure planning, Tableau is used to analyze traffic density, land-use patterns, and public utility distribution, supporting informed planning decisions. Business and logistics applications include regional sales analysis, market penetration studies, and optimization of distribution routes based on geographic demand. Additionally, in public health and governance, Tableau assists in mapping disease incidence, healthcare facility accessibility, and population demographics, helping policymakers identify underserved areas. Advanced functionalities such as layered maps, spatial joins, custom territories, and interactive dashboards allow multiple datasets to be overlaid and explored dynamically, thereby enhancing spatial understanding and supporting data-driven decision-making across environmental, commercial, and societal domains.At the end of

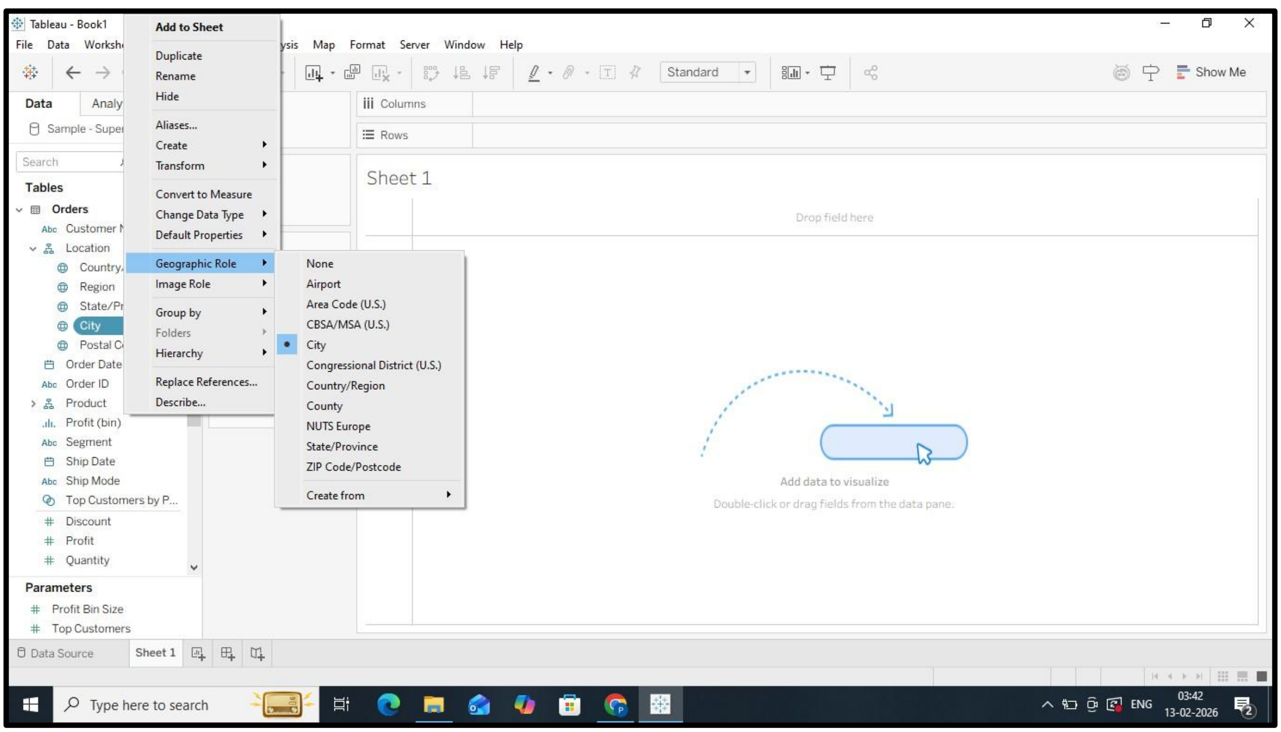
this experiment we can conclude that geospatial data was successfully visualized using multiple map representations in Tableau. Spatial patterns, regional variations, and significant trends were identified and interpreted, demonstrating the effectiveness of Tableau for location-based data analysis.

Tasks to be implemented

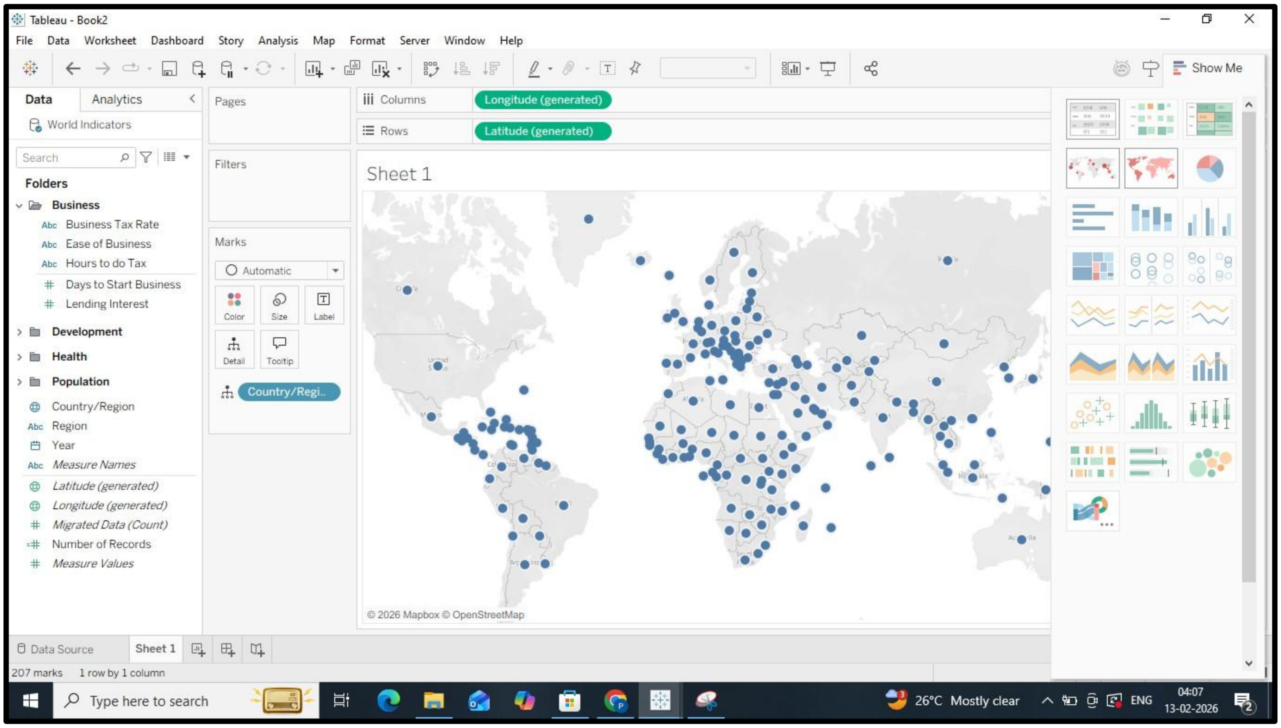
TASK 1: Observation after Plotting Data (Basic Geospatial Plot)

Steps in Tableau

1. **Load the dataset**
   - Open Tableau → *Connect* → select the data source (Excel/CSV/Database).
   - Verify that geographic fields (Country, State, City, Latitude, Longitude) are correctly recognized (globe icon).
2. **Assign geographic roles**
   - Right-click the location field → *Geographic Role* → select the appropriate role (e.g., State, City).



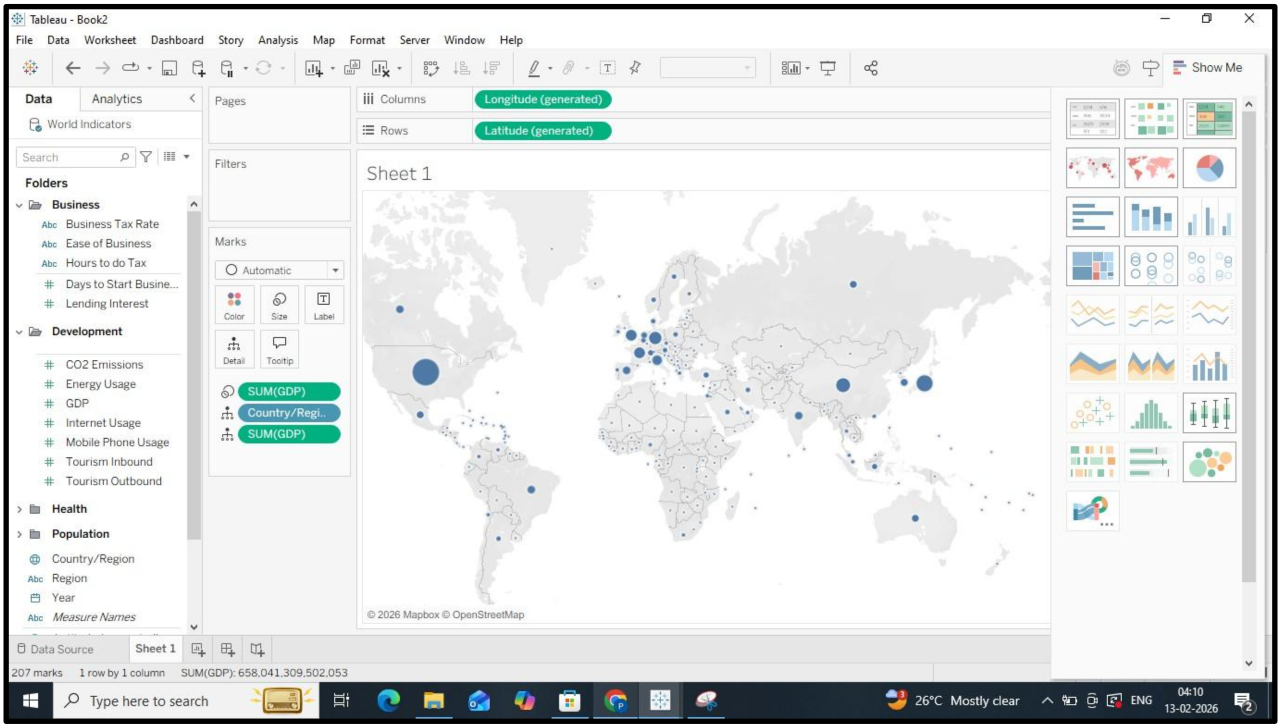
3. **Create a basic map**
   - Drag the geographic field to the *Rows* or *Columns* shelf, or directly to the canvas.

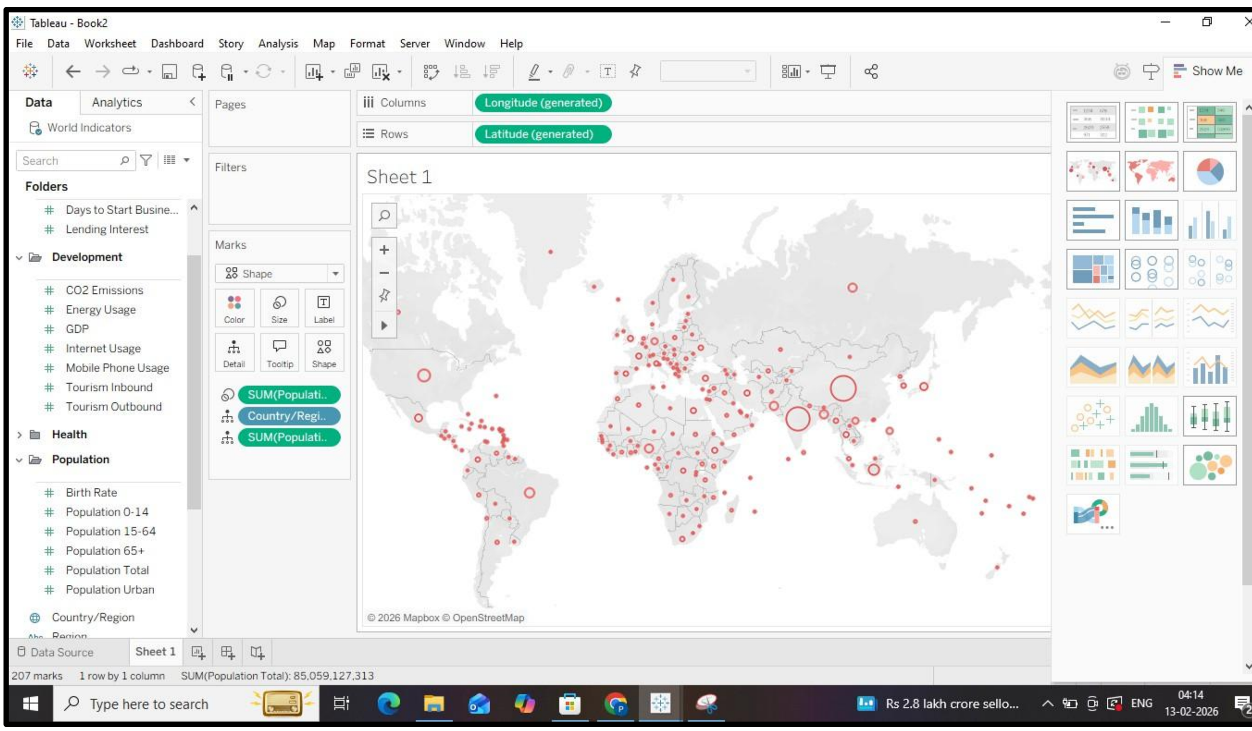


- Tableau automatically generates a map view.

4. **Plot measurement data**

   - Drag the relevant measure (e.g., water quality parameter, sales value, population) to *Color* or *Size* on the *Marks* card.





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5. **Adjust visualization settings**

- Use the *Marks* card to change mark type (circle, filled map).
- Adjust color gradients and size for better visibility.

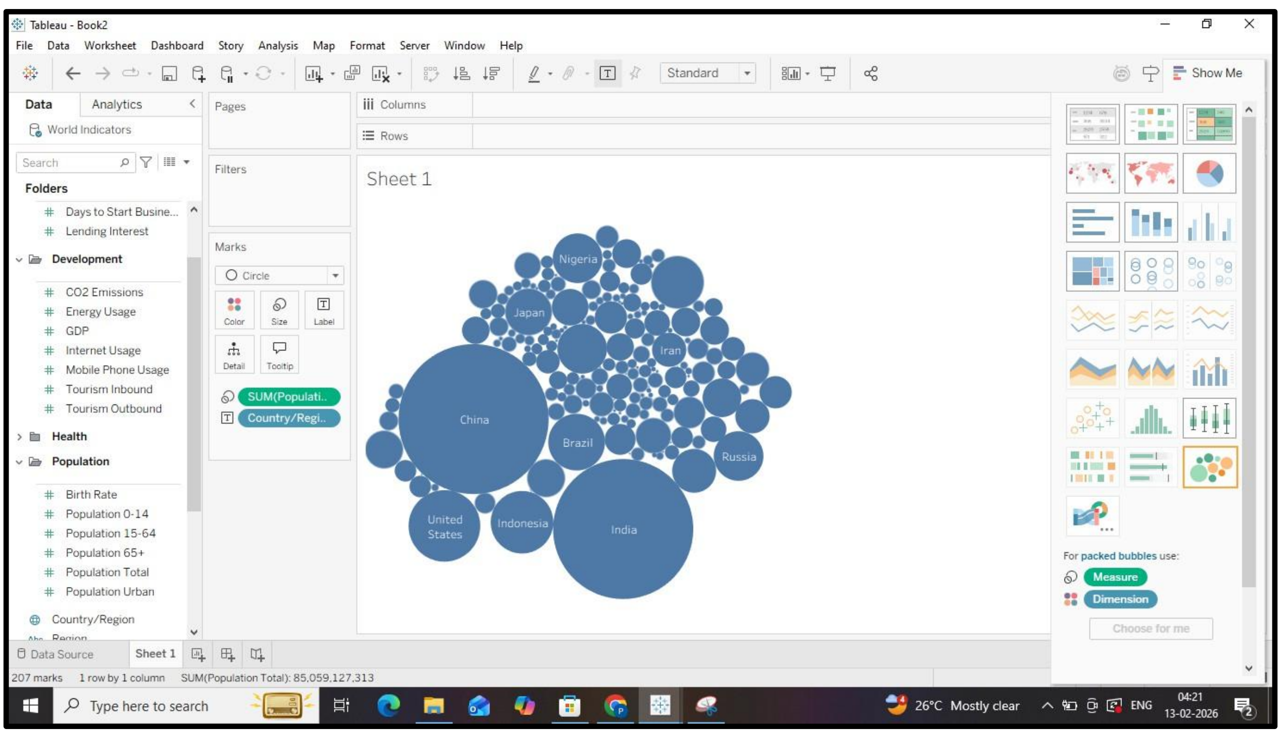
Observation

- Identify spatial distribution patterns such as clustering, or regional dominance.
- Observe variations in values across locations and detect any outliers or anomalies.
- Note initial trends such as high or low concentration zones.

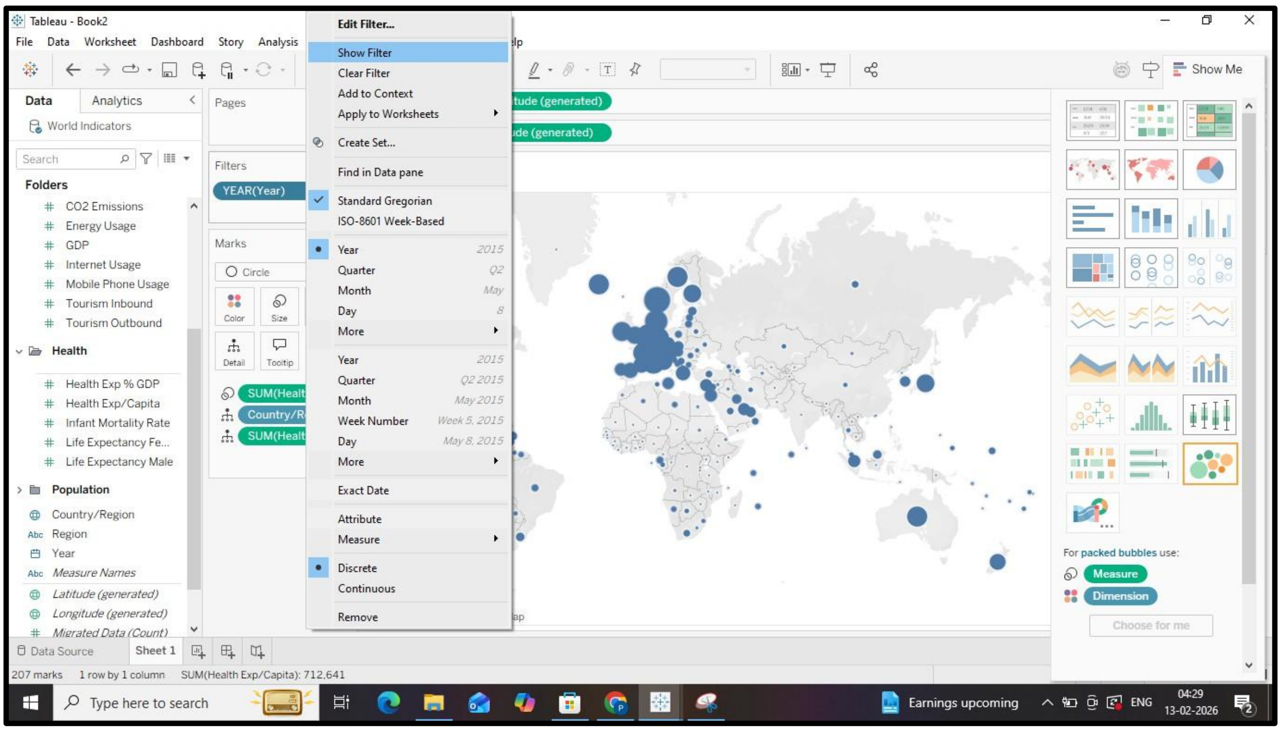
TASK 2: Observation After Plotting Various Forms of Maps (Based on Visualization Choices)

Steps in Tableau

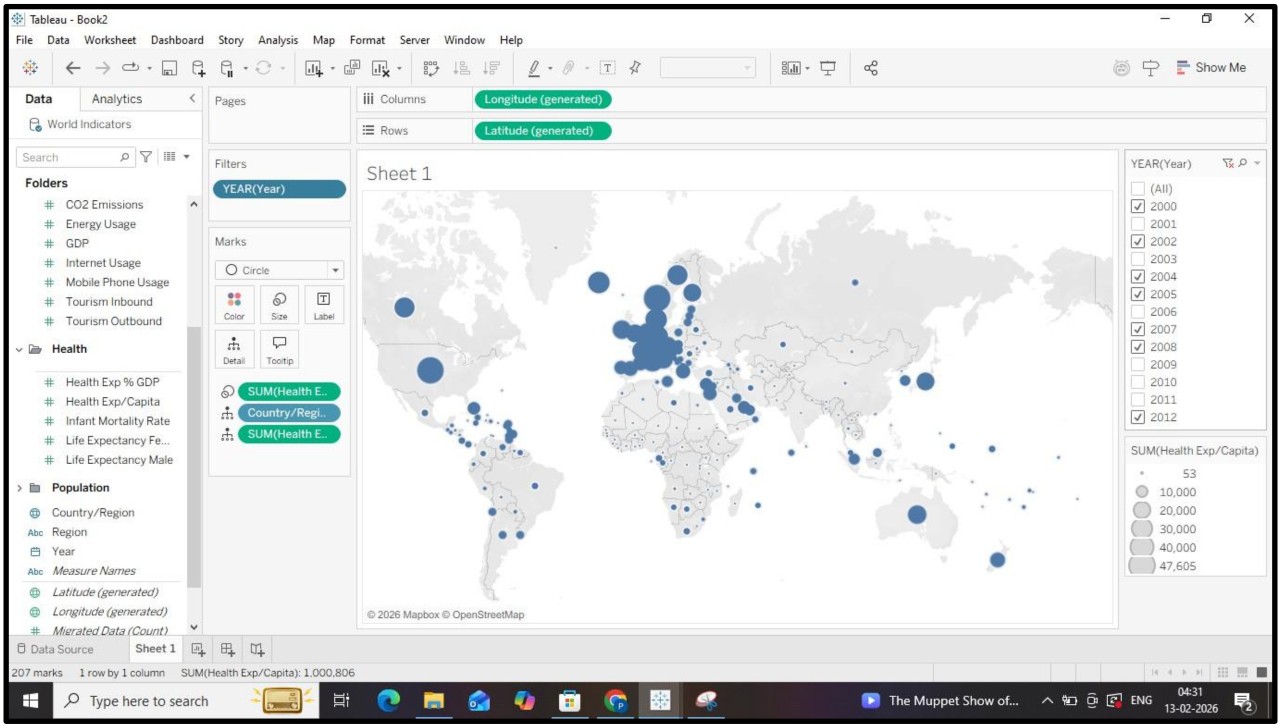
1. **Switch between map types**
   - From the *Marks* card, change the mark type to:
     - *Symbol Map* for point-based data
     - *Filled Map* for area-based comparison
     - *Density Map* to identify hotspots



2. **Apply multiple measures**
    - Drag additional measures to *Color*, *Size*, or *Label* to represent different attributes.
    - Use *Measure Names* and *Measure Values* to compare multiple parameters.
3. **Use layers and map details**
    - Enable map layers such as borders, streets, and landmarks from *Map* → *Map Layers*.
    - Overlay different datasets if available using spatial joins or blending.
4. **Apply filters**
    - Add filters based on time, category, or threshold values to refine the map.



- Use interactive filters to dynamically update the visualization.



5. **Customize tooltips**
    - Edit tooltips to display contextual information when hovering over regions or points.

   Observation

- Compare how different visualization types emphasize different spatial characteristics.
- Identify hotspots, gradients, and regional contrasts more clearly.
- Observe how filtering and layering reveal hidden spatial relationships.

TASK 3: Interpretation of the Visualized Map

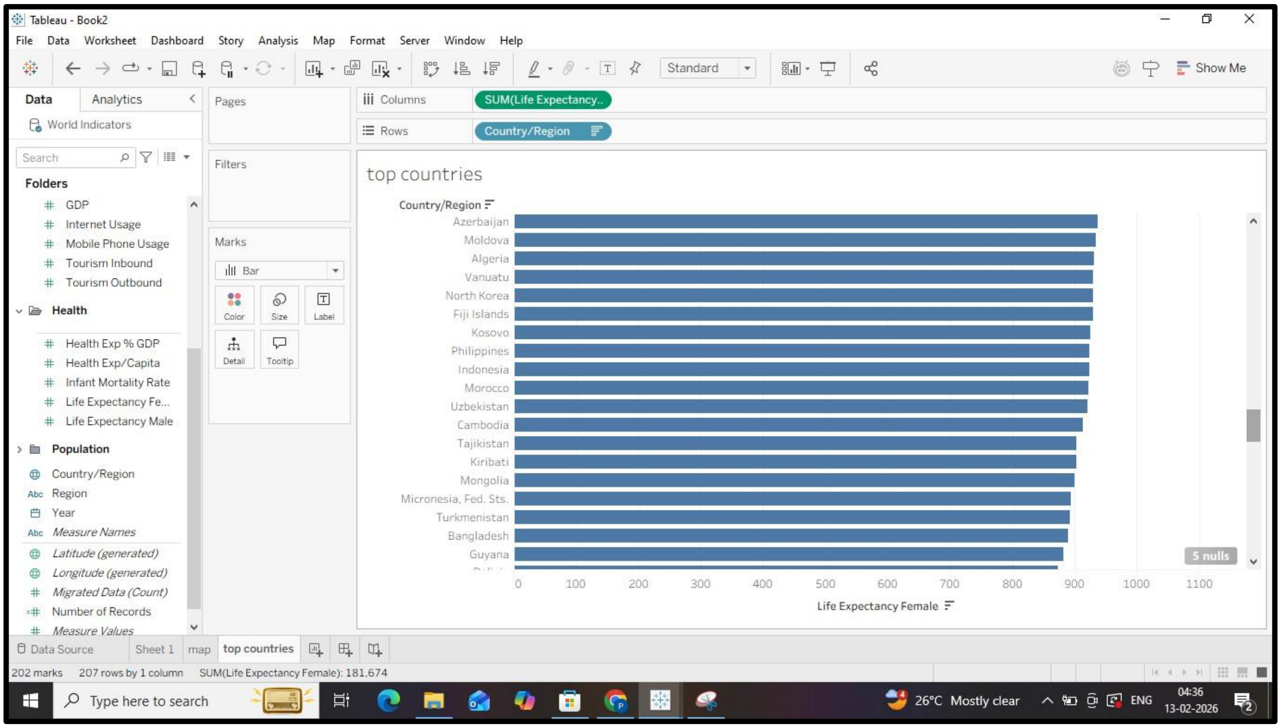
Steps in Tableau

1. **Analyze spatial patterns**
    - Examine color intensity, symbol size, and density to understand magnitude and variation.
    - Identify consistent high-value or low-value regions.
2. **Compare across regions**
    - Use legends and scale bars to interpret relative differences.

- Analyze regional disparities and spatial correlations.

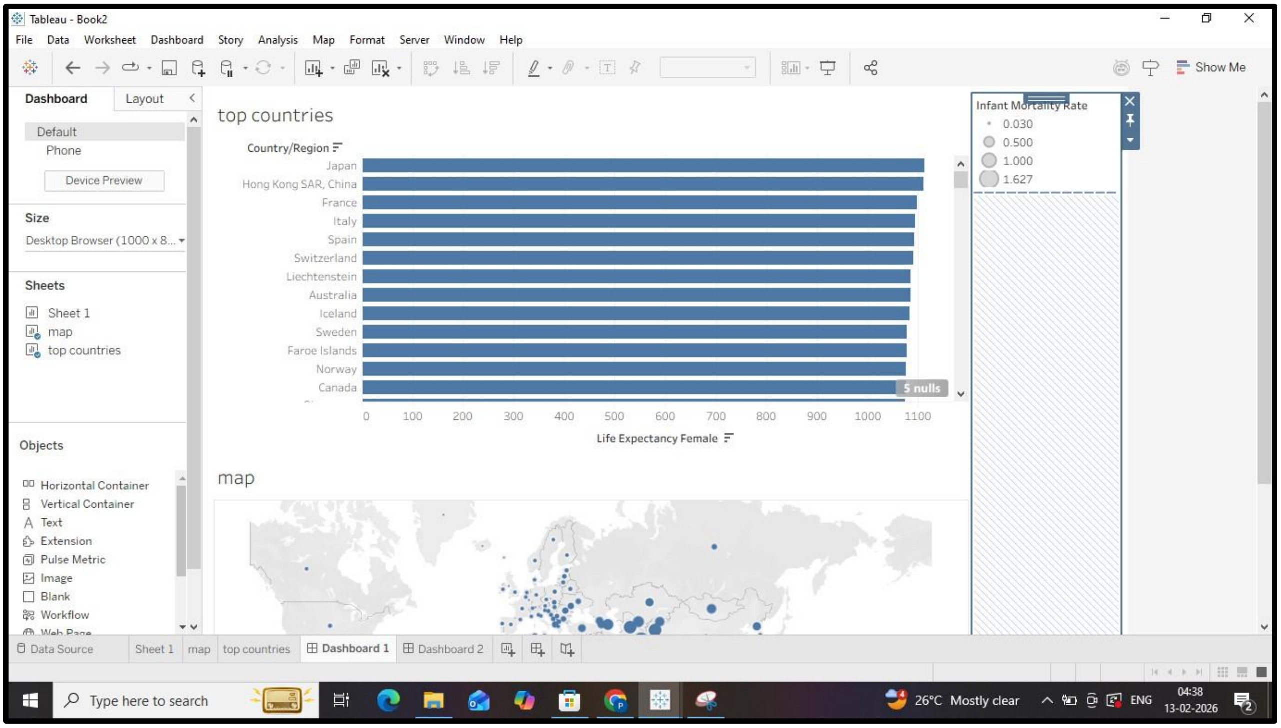
3. **Use interactivity for deeper insight**

   - Hover over points or regions to read detailed tooltips.
   - Use filters and parameters to analyze changes across time or categories.



4. **Create dashboards**

   - Combine maps with charts, tables, or KPIs in a dashboard.
   - Enable actions such as highlight, filter, or drill-down for interactive analysis.



5. **Draw conclusions**

    - Relate observed spatial trends to real-world factors such as geography, infrastructure, population, or environmental conditions.

    - Document insights that support decision-making or further analysis.

Interpretation

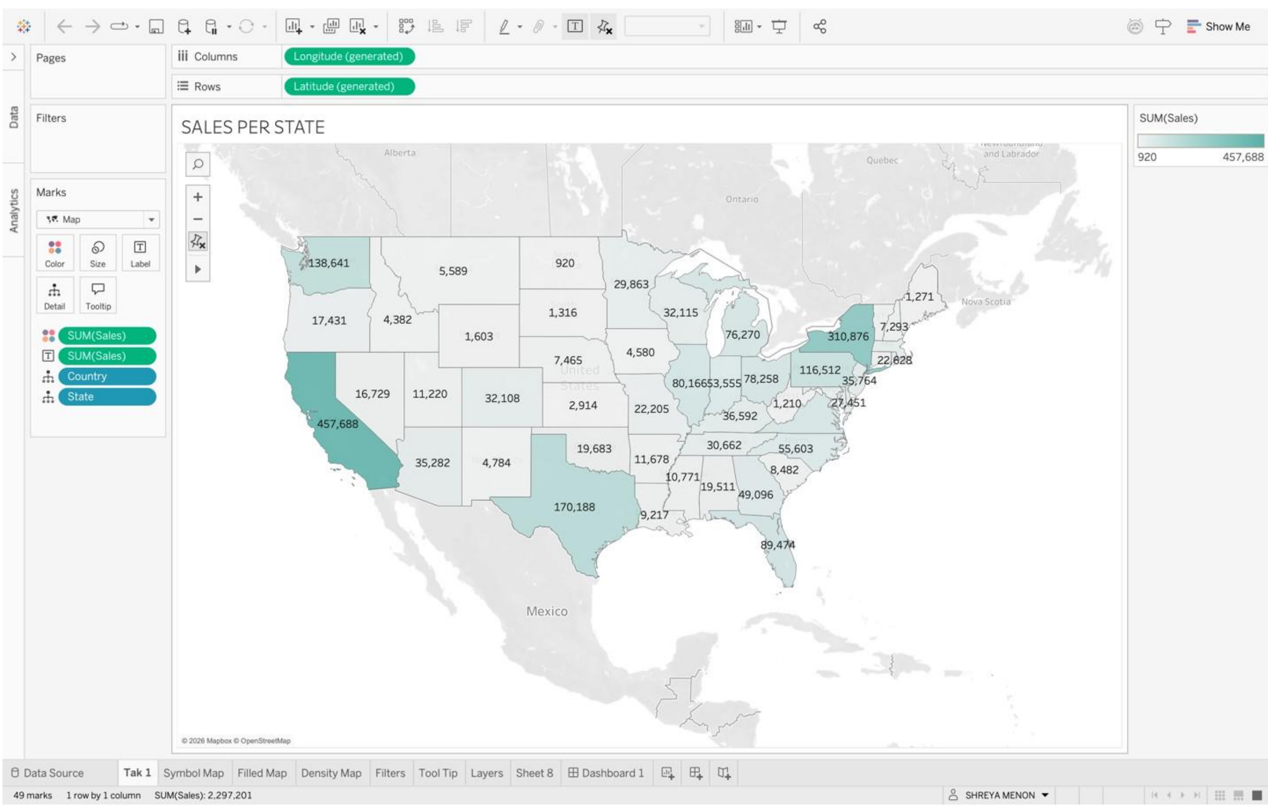
- Translate visual patterns into meaningful insights.
- Explain why certain regions show higher or lower values.
- Support conclusions using visual evidence from the map

Sample task to be observed in the Laboratory

1. Search/locate and download the geospatial Data (Use same dataset if it contains location information)
2. To learn how to visualize geospatial data
    a. Auto Geo-tagging
    b. Custom Geo-tagging
3. Apply heat map
4. Try various forms of heat maps
5. Analyse the visualization and write your interpretation after observation on heat-map

6. Interactive filtering over map
7. Following maps should be demonstrated
    a. Proportional symbol maps
    b. Choropleth maps (filled maps)
    c. Point distribution maps
    d. Density maps (heatmaps)
    e. Flow maps (path maps)
    f. Spider maps (origin-destination maps)

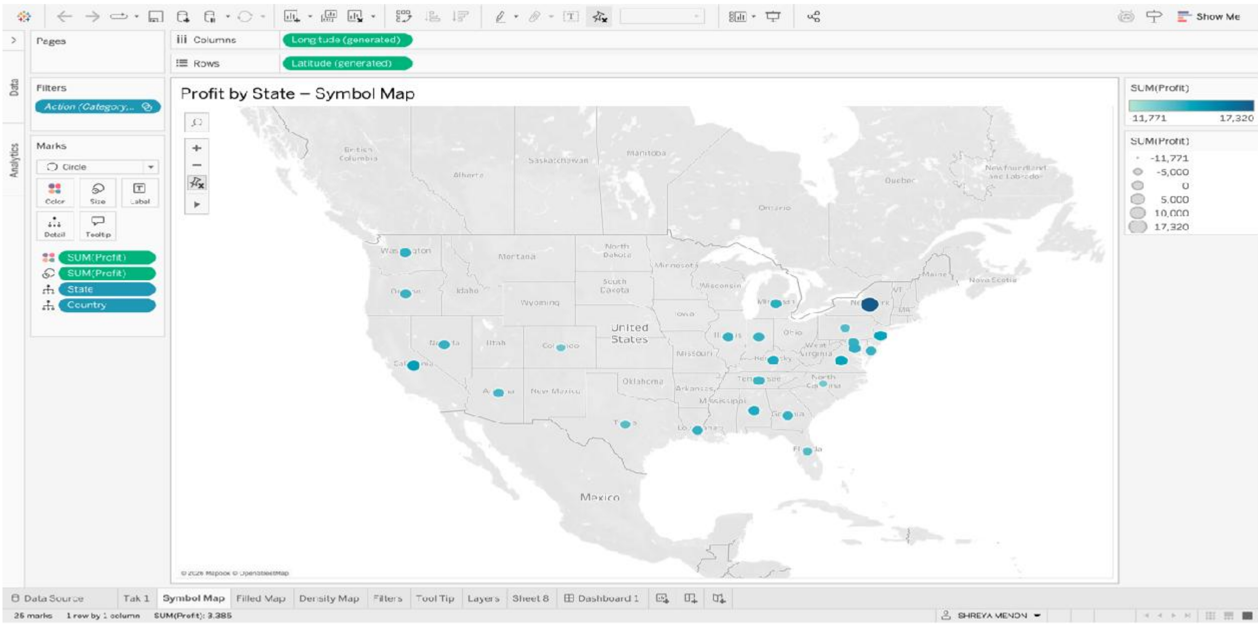
<u>TASK 1: Observation after Plotting Data (Basic Geospatial Plot)</u>



The above plotted map shows the sales per state – California clocking the highest sales of all states.

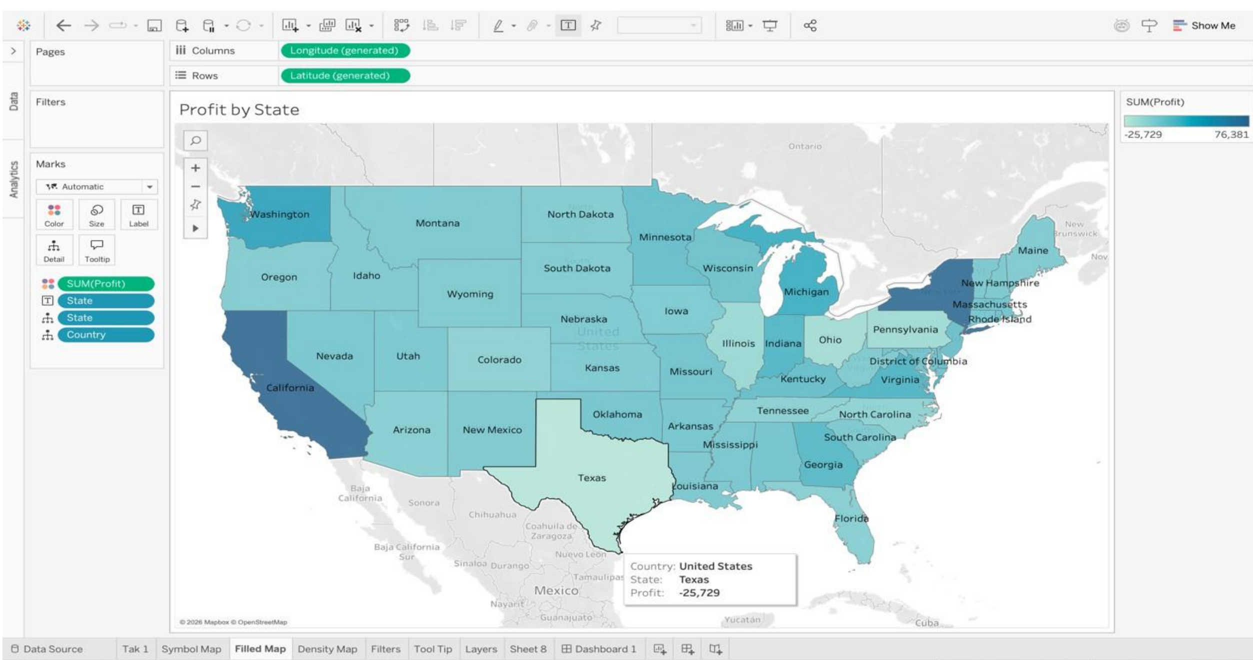
TASK 2

Symbol Map



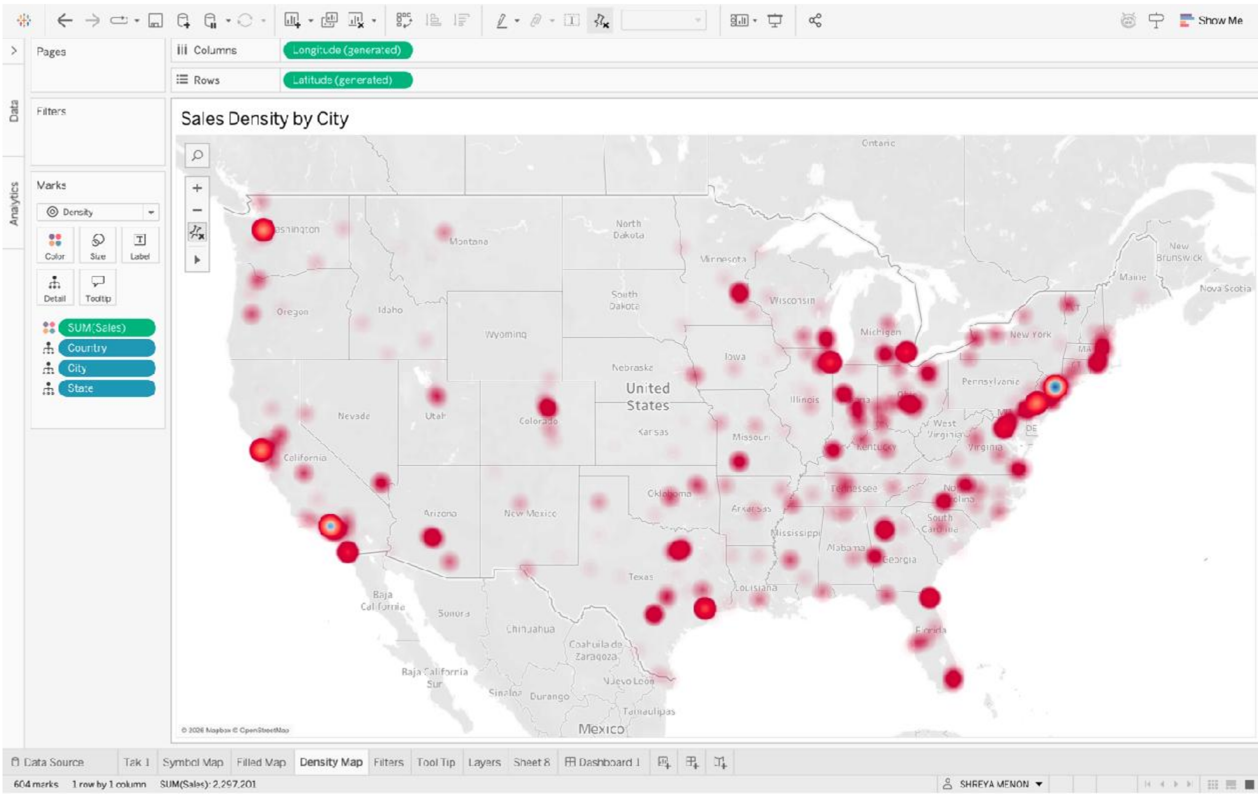
The size and colour of the points shows the magnitude of Profit – with New York showing the max profit with the biggest and darkest point.

Filled Map



The colour gradient display the Profit by State the lighter states reporting lower profits to the darker shades showing more profit.

Density Map



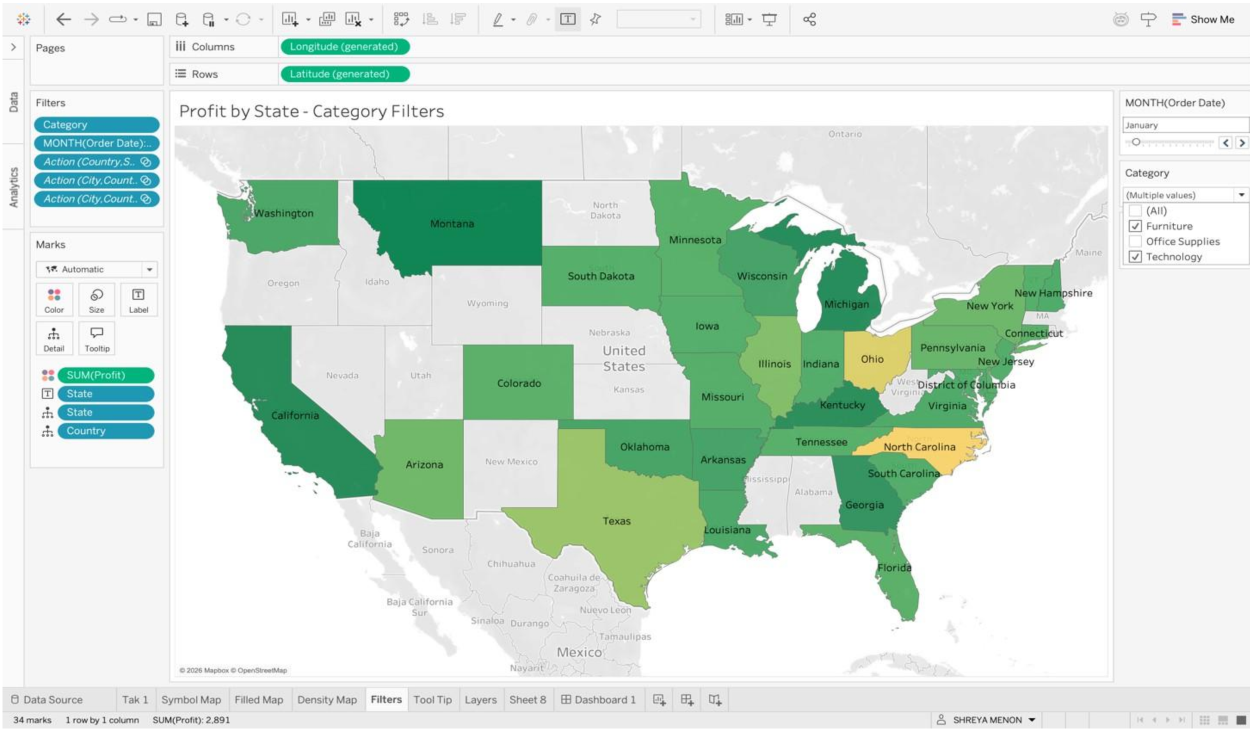
The above visualization shows sales density across the cities in each state with more sales towards the East Coast of the USA

Filter Map

Map showing profit by state with filters of :

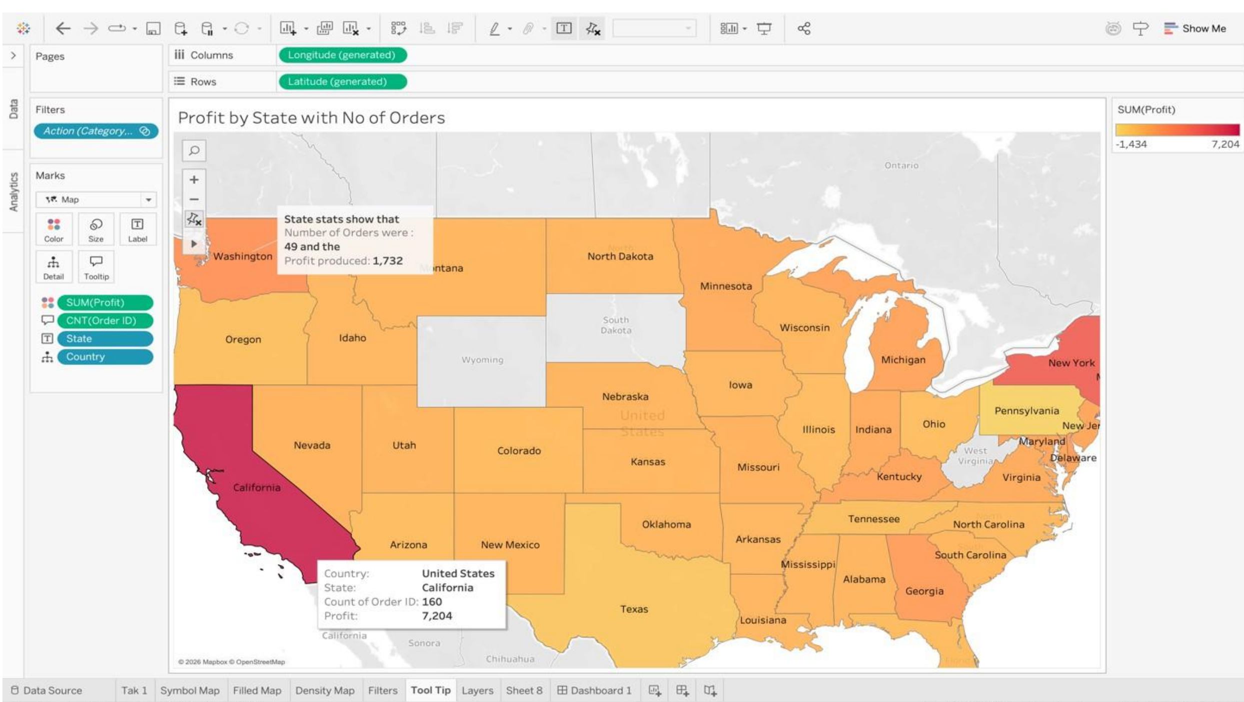
1) Category of Products

2) Month of the year



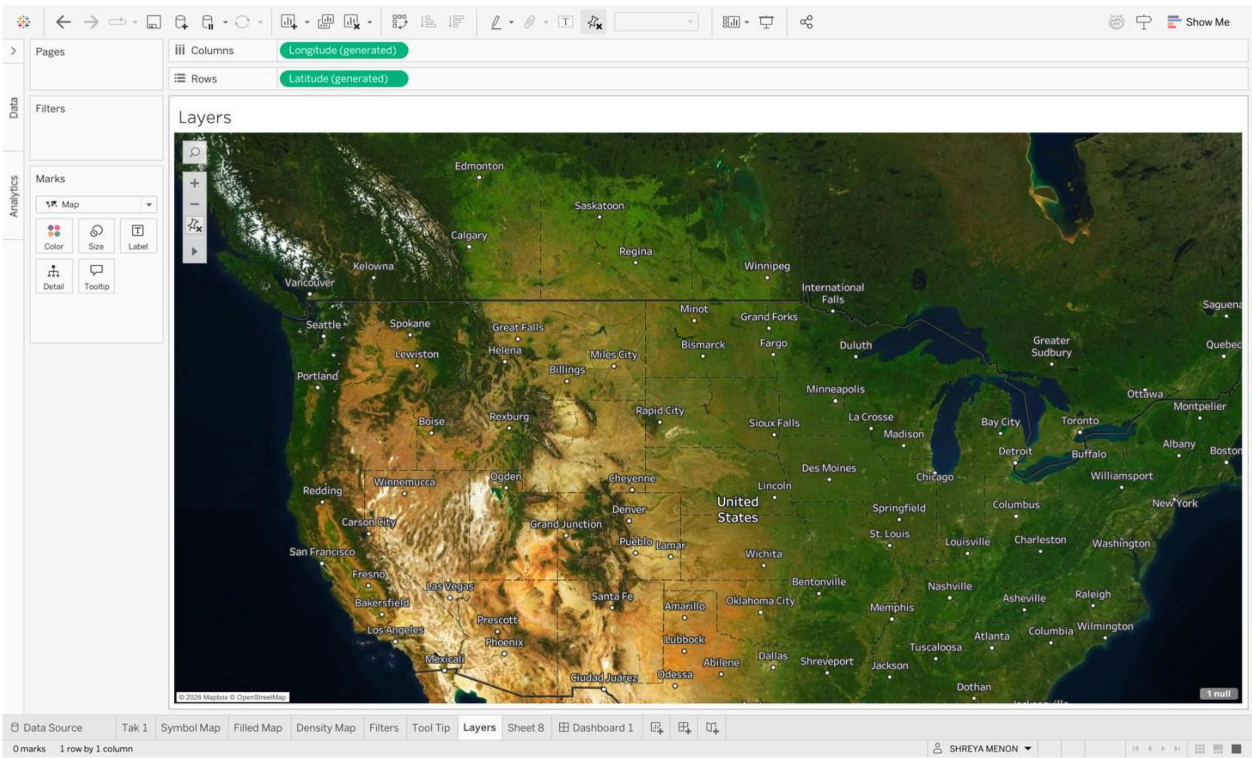
The above visualisation shows that Montana and California are amongst the states to report maximum profit in the month of January – in the technology, furniture category of products.

<u>Tool tip Customization</u>



Visualization shows the profit recorded by each state along with the tool tip customization where in the number of orders per state is also showcased.

Map Layers

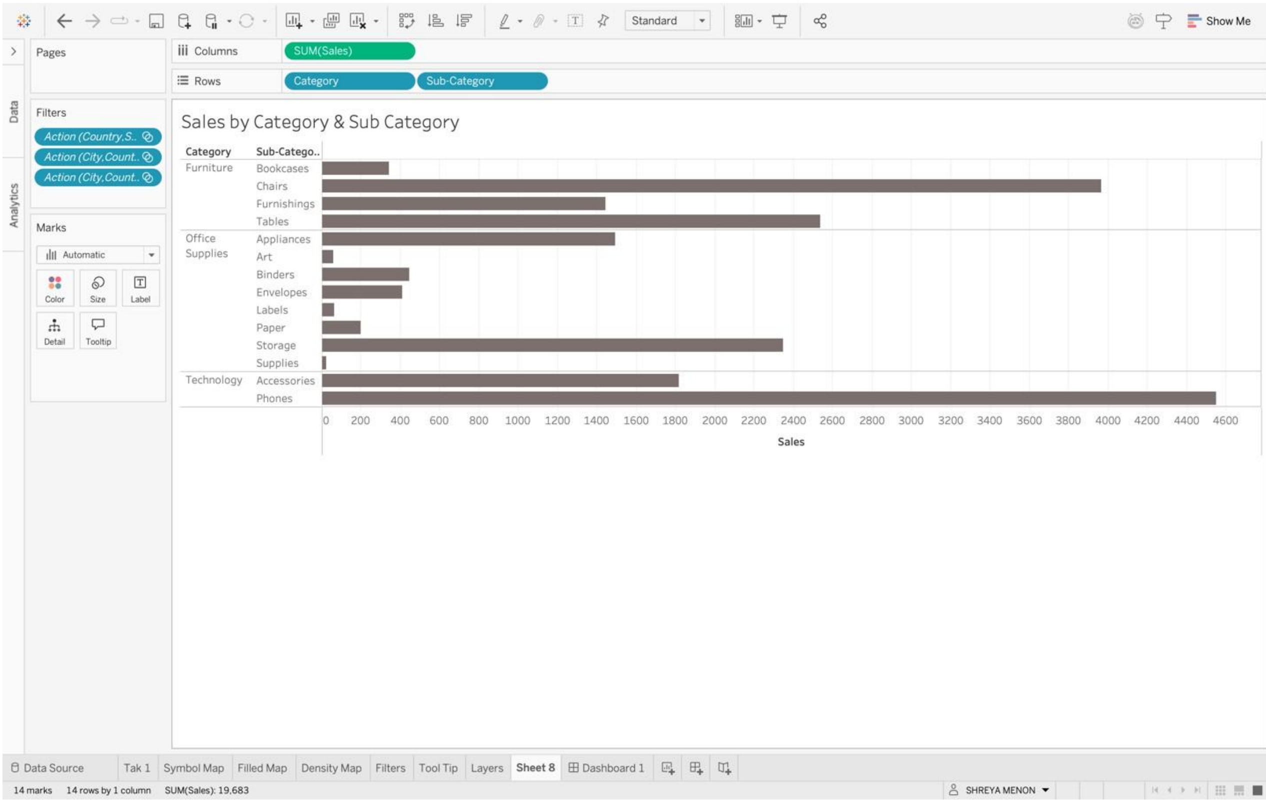


Provides satellite view of the map.

Task 3:

Bar Chart

Provides sales by Category and Sub-Category



Dashboard

This provides the bar chart along with 3 major visualizations of Order Month and Category

1) Sales Density by City

2) Sales by Category & Sub-Category

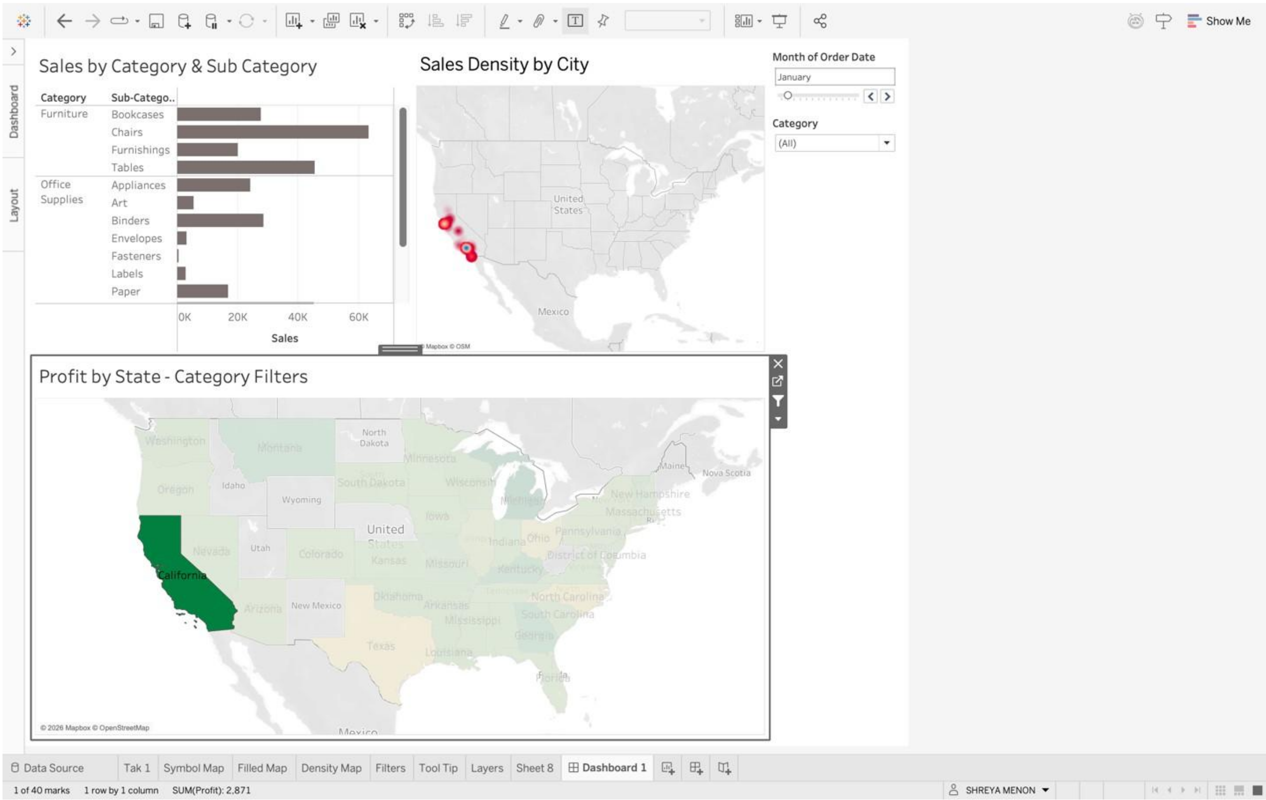
3) Profit by State

In the dashboard below we see for the state of California amount of sales across the categories and sub-categories with the help of the bar chart.

The density map also shows the density of Sales across the cities in California.

The filled map shows that the state in of the highest profit makers due to the darker colour gradient.

This statistics for the month of January but across all categories.



Post Laboratory Questions:

1.Explain the Choropleth maps

A **choropleth map** is a thematic map in which different geographic regions (such as states or districts) are shaded or colored according to the value of a specific variable. The color intensity represents the magnitude of the data — darker or deeper shades usually indicate higher values, while lighter shades indicate lower values.

Choropleth maps are commonly used to display statistical data such as population density, literacy rate, pollution levels, or income distribution across regions. They help in identifying spatial patterns, trends, and regional differences easily.

For example, in water quality analysis, states with higher BOD values can be shown in darker shades to indicate higher pollution levels.